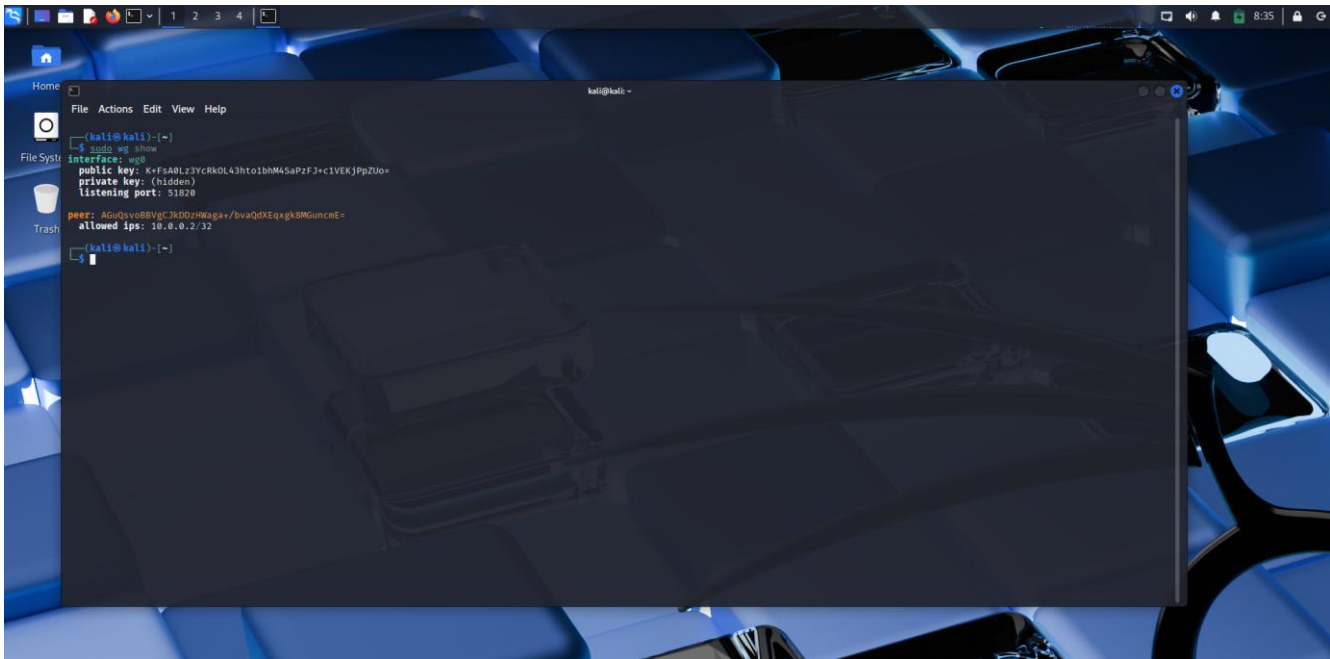


Lab Activity: Deploying Modern VPNs with WireGuard



COMPARATIVE PING LATENCY (ms)

VPN PROTOCOL	PING (ms)
 OPENVPN	 58.5 ms ▲
 WIREGUARD	 24.2 ms ▼

1. Attack Surface: Compare the lines of code between OpenVPN and WireGuard. Why does a smaller codebase correlate with better security?

OpenVPN relies on a massive codebase of hundreds of thousands of lines of code, whereas WireGuard is intentionally designed to be extremely lean, typically consisting of only around 4,000 lines. A smaller codebase directly correlates with better security because it significantly reduces the "attack surface." With fewer lines of code, there are simply fewer opportunities for hidden software

bugs, memory leaks, and exploitable vulnerabilities to exist. Additionally, a compact codebase can be thoroughly and comprehensively audited by security researchers in a short amount of time, whereas auditing OpenVPN in its entirety is incredibly difficult.

2. Stealth Mode: WireGuard does not respond to packets unless they are correctly encrypted with a known public key. How does this protect the gateway from "port scanning"?

Traditional VPN protocols often reply to unauthenticated incoming packets with an error or rejection message. This normal networking behavior inadvertently tells an attacker running a port scanner that a VPN service is actively listening on that port. Because WireGuard will not respond unless a packet is correctly encrypted with a known public key, it silently drops unauthorized traffic without sending any reply. To an attacker's port scanner, the server appears completely unresponsive and the port looks closed, effectively hiding your gateway from automated network sweeps.

3. Roaming: Why does a WireGuard connection stay "alive" when a someone moves their laptop from a home network to a coffee shop, while OpenVPN often requires a reconnect?

OpenVPN is a stateful protocol that binds an active session to a specific IP address. When a user switches networks (like moving from home Wi-Fi to a coffee shop or 5G), their external IP address changes, causing OpenVPN to drop the session and force a reconnect. WireGuard, conversely, is stateless. It utilizes "Cryptokey Routing", meaning it authenticates the connection based on the user's cryptographic identity (their public key) rather than their location. When a user changes networks, the next data packet sent is still encrypted with their private key. The gateway receives it, safely decrypts it using the known public key, and seamlessly updates the client's new external IP address on the fly without ever dropping the tunnel.

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